

RNA-seq Bioinformatics Project

Early Detection Biomarkers for Infection Using RNA-seq Transcriptomics

A complete bioinformatics pipeline to discover molecular signatures distinguishing healthy individuals from infected patients — enabling early detection before clinical symptoms appear.

10,346

Genes

397

Samples

4

Infection Stages

20

Biomarkers

Project Details

Author: Esraa Ramadan Mohamed Ahmed

Type: Voluntary Research Project

Tools Used: Python, T-test, PCA, Heatmap, Venn Diagram

Dataset: Diabetes & Infection RNA-seq Dataset (10,346 genes × 397 samples)

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Analysis Pipeline — 7 Steps

01

Data Loading & Quality Control

Loaded 10,346 genes × 397 samples. Fixed naming mismatch (dots → dashes). Checked for missing values and confirmed metadata alignment.

02

First Filtration — Sparsity Removal

Total counts: 1,221,428 | Sparsity: 29.74%. Removed sparse genes (zero-based filtering at 20% threshold).

Raw genes: 10,346 → After filtration: 10,119

03

Sample Distribution & Second Filtration

4 groups created: Healthy (336), Early (14), Middle (29), Late (18).

Second filtration: mean expression > 10 counts in at least one infection stage.

Genes: 10,119 → 3,262 (prevents healthy group from biasing the filter)

04

CPM Normalization & Log2 Transformation

CPM corrects for library size differences. $\text{Log}_2(\text{CPM}+1)$ stabilizes variance and produces a near-normal bell curve centered at $\text{Log}_2 \approx 7-8$, making statistical tests valid.

05

PCA — Global Transcriptomic Separation

PC1 (25.70%): Healthy vs. Infected axis.

PC2 (13.94%): Secondary immune variation.

Healthy cluster = right side. Infected samples shift LEFT along PC1 as disease progresses.

06

T-test DEGs — Differential Expression

Applied T-test (Healthy vs. Each Stage) with $p < 0.05$ and $|\text{Log}_2\text{FC}| > 0.58$.

Early: 82 DEGs | Middle: 142 DEGs | Late: 22 DEGs

07

Top 20 Biomarkers — Heatmap & Clustering

Scored genes by Z-score, p-value, and Log_2FC . Identified 20 early biomarkers split into:

Steady Markers: Always ON across all stages.

Switching Markers: High in Early, drop in Late.

Key Results & Findings

Venn Diagram — T-test vs DESeq2 Cross-Validation

Stage	Total DEGs	Unique	Shared
Early	82	69	Early+Middle: 6 Early+Late: 6
Middle	142	133	Middle+Late: 2
Late	22	13	All 3 stages: 1 gene

Top 20 Early Biomarkers

Marker Type	Genes	Biological Meaning
Steady Markers (Always ON)	LOC100129697, POLE4, KLF13, TCP11L2, UBL4A	Maintained high expression across ALL infection stages — universal diagnostic anchors
Switching Markers (Molecular Timers)	FOS, GLB1L2	HIGH in Early infection → LOW in Late. Act as molecular timers to determine disease stage
Special Case	THAP7	High → drops at Middle → rises at Late. Suggests biological reset during peak immune activity

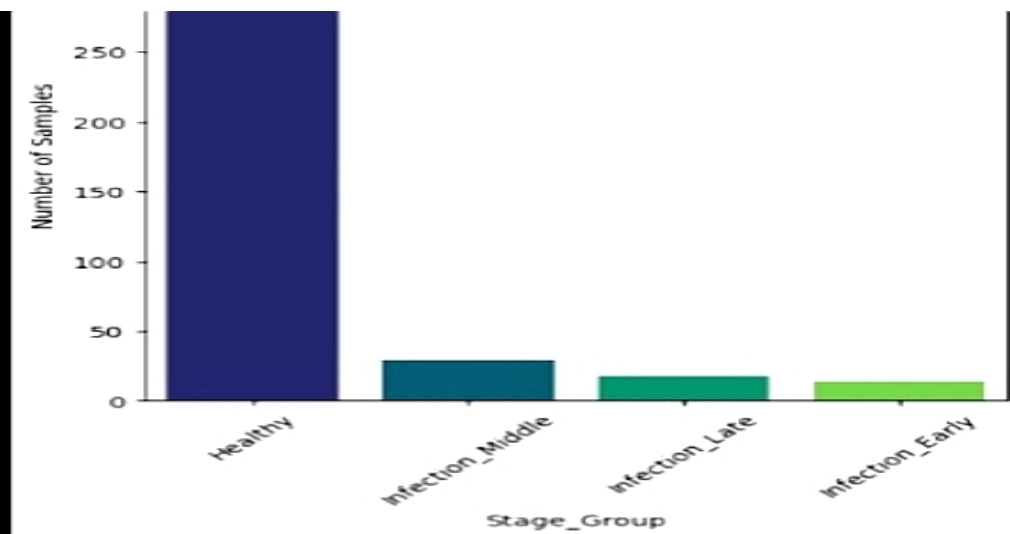
Conclusions & Future Directions

- ✓ A robust 20-gene expression signature was identified through the T-test pipeline.
- ✓ The Dual-Layer approach (Steady + Switching markers) enables precise molecular staging of disease progression.
- ✓ Steady markers confirm infection presence at every stage.
- ✓ Switching markers (FOS, GLB1L2) determine both IF infection is present AND WHEN it started.

Future Work:

- Clinical validation of the 20-gene signature on independent patient cohorts.
- Transition to DESeq2/EdgeR for rigorous RNA-seq dispersion modeling.
- Develop ML classifiers (Random Forest, SVM) for automated infection staging.

Result 1: Dataset Overview — Sample Distribution per Stage



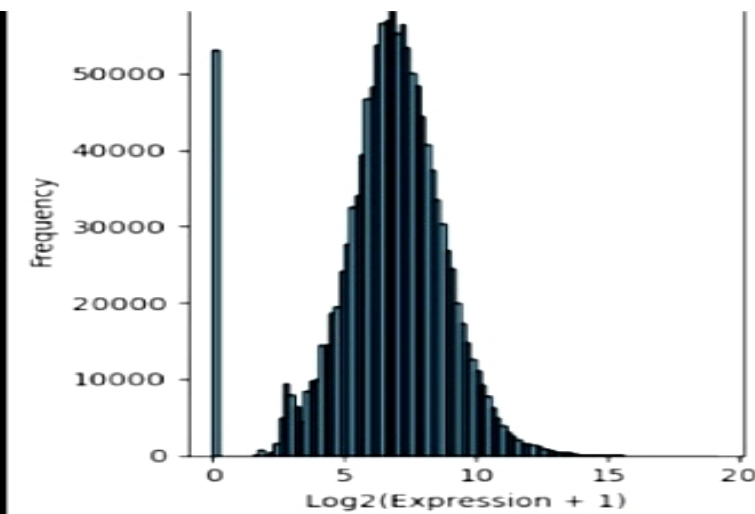
Commentary

Key Observation: The dataset is heavily imbalanced. The Healthy group dominates with ~336 samples; all three infection stages combined total only ~61 samples (Early ~14, Middle ~29, Late ~18).

Why it matters: This reflects reality — healthy patients outnumber infected ones. The T-test must compare these unequal groups carefully. The large Healthy baseline gives strong statistical power for defining 'normal' expression.

Conclusion: The Healthy group is our robust reference baseline. Infection_Early is the smallest and hardest to detect — yet the most clinically valuable target.

Result 2: CPM & Log2 Normalization — Stabilizing the Data



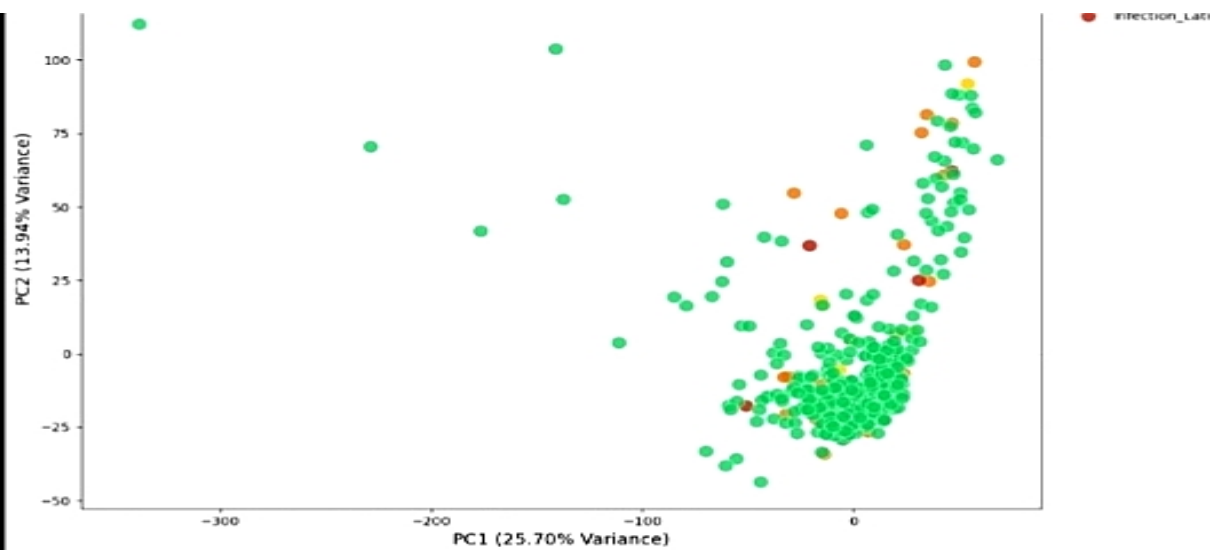
Commentary

Why CPM Normalization? Raw RNA-seq counts are biased by sequencing depth (library size). Samples sequenced more deeply appear to express every gene more highly — an artifact. CPM corrects this by scaling all samples to the same depth.

Why Log2 Transformation? Before normalization the distribution is heavily skewed — a few 'giant' genes dominate. Log2 compresses extreme values, converting multiplicative fold-changes into additive differences.

What the histogram shows: After $\text{Log}_2(\text{CPM}+1)$, the distribution follows a near-normal bell curve centred around $\text{Log}_2 \approx 7-8$. This makes T-tests and PCA statistically valid and prevents outlier genes from dominating results.

Result 3: PCA — Global Transcriptomic Separation by Infection Stage



Commentary

PC1 (25.70%): Captures the primary axis — healthy vs. infected.

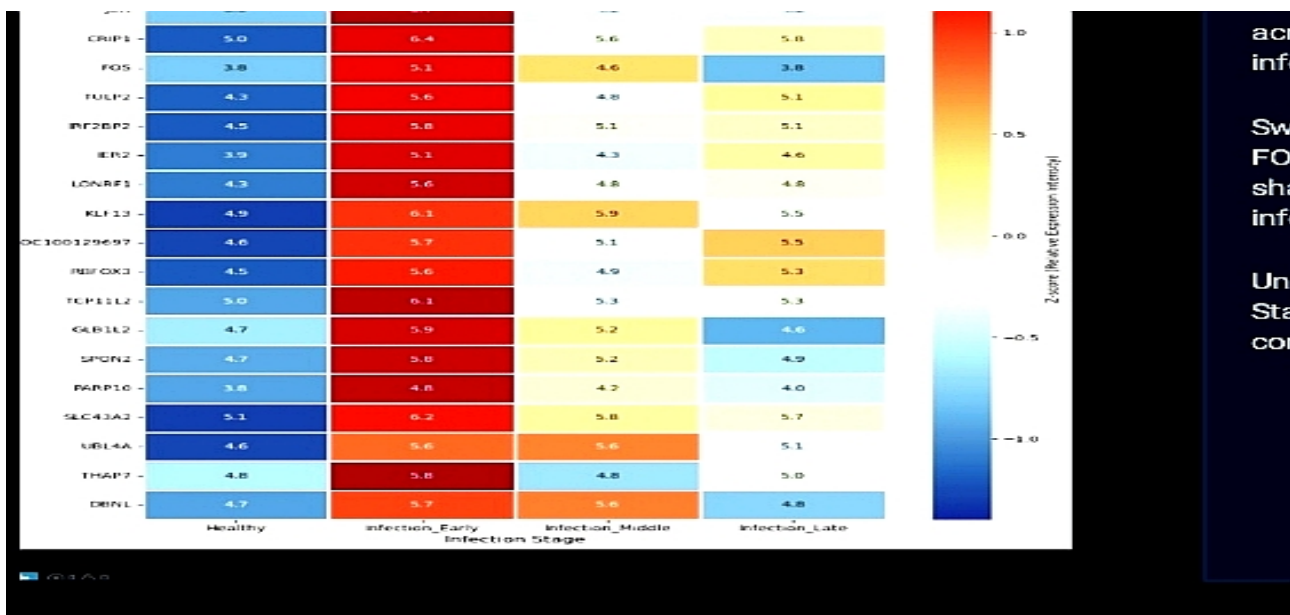
PC2 (13.94%): Captures secondary immune variation.

Healthy (green): Dense cluster on the right — stable homeostatic baseline.

Disease Trajectory: Samples shift LEFT along PC1 as infection progresses (Early → Middle).
Late-stage samples begin returning toward the healthy cluster as the immune response resolves.

Conclusion: The partial overlap between Healthy and Early stages justifies using T-test to isolate specific genes that global PCA cannot fully separate.

Result 4: Top 20 Biomarkers — Temporal Expression Heatmap



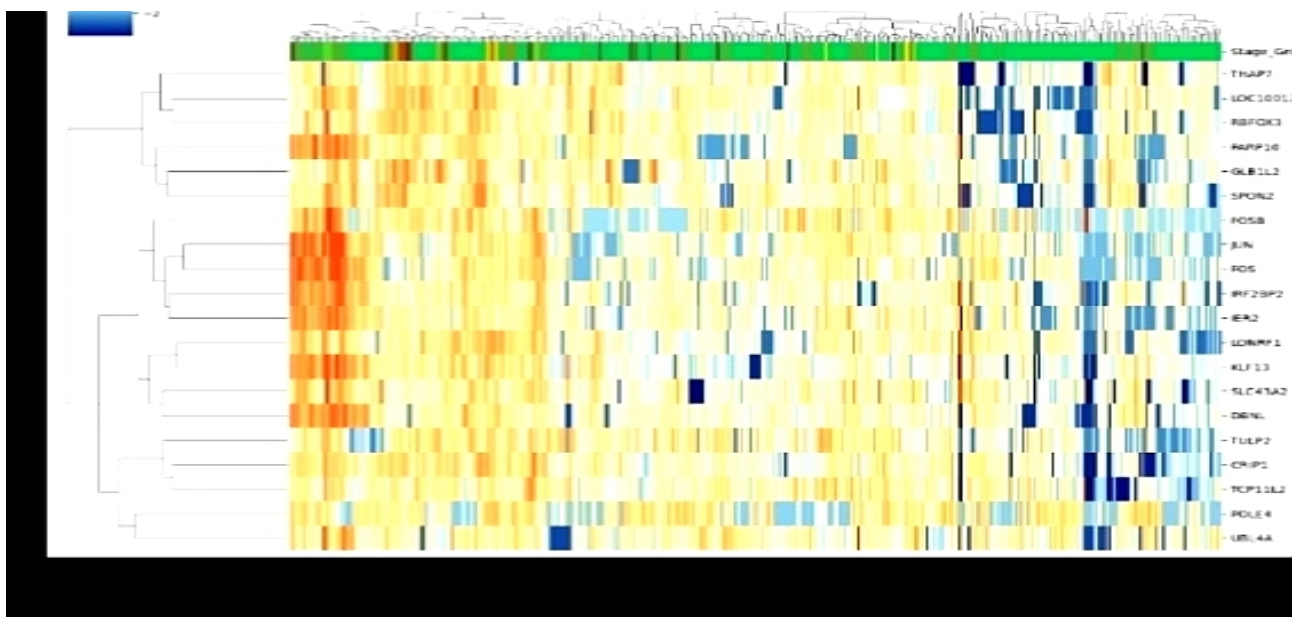
Commentary

Steady Markers — 'Always On': LOC100129697, POLE4, KLF13 maintain high expression (orange/red) across ALL infection stages. They act as constant alarms confirming infection regardless of timing.

Switching Markers — Molecular Timers: FOS and GLB1L2 surge during Early infection (deep red) then drop sharply at Late stage (blue). If FOS is HIGH → Early infection; if LOW → infection has progressed.

Unique Case — THAP7: Starts high → drops at Middle → rises again at Late. Suggests a complex biological reset during peak immune activity.

Result 5: Hierarchical Clustering — Molecular Fingerprint of Infection



Commentary

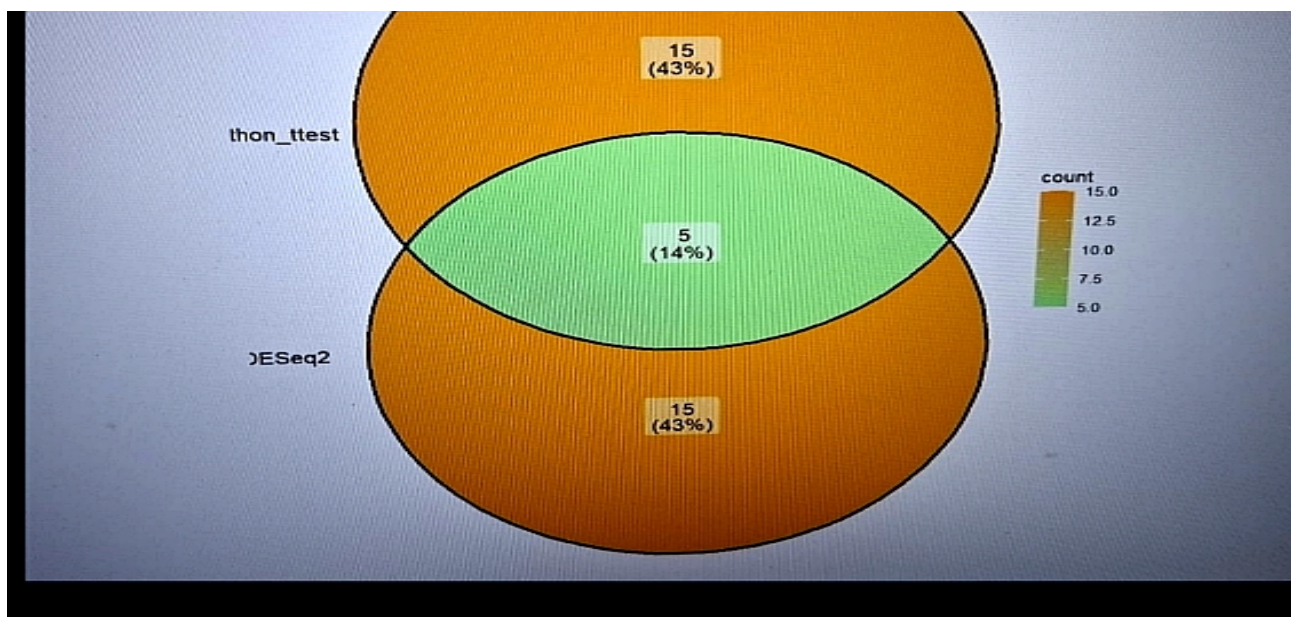
High-Resolution Patient Map: Each column = 1 patient; each row = 1 biomarker gene. The dendrogram groups similar patients and genes.

The Infected 'Red Block' (left): A clear block of high expression (red/orange) dominates the left — mostly Early & Middle patients. This is the infection's molecular fingerprint.

Healthy Cluster (right): Distinct blue-white zone — low baseline expression consistent with homeostasis.

Steady Markers: TCP11L2, POLE4, and UBL4A maintain an orange-red band across ALL infected samples — confirmed universal diagnostic anchors.

Result 6: Venn Diagram — T-test vs DESeq2 Cross-Validation



Commentary

What this shows: A comparison of differentially expressed genes identified by two independent methods: the Python T-test pipeline and DESeq2 (rigorous R-based RNA-seq analysis).

Results:

- 15 genes (43%) — T-test only
- 15 genes (43%) — DESeq2 only
- 5 genes (14%) — shared by BOTH

The 5 Shared Genes: NCAPH2, JUNB, TGFB1, DLG4, KLF13 — validated by two independent methods; highest confidence biomarker candidates.

Conclusion: T-test provides a useful first screen. DESeq2 offers more accurate dispersion modeling. The 5 shared genes are the most reliable targets for clinical validation.

Python Analysis Code — Key Steps

Phase 1 — Data Loading & Quality Control

```
import pandas as pd
import numpy as np
from scipy.stats import ttest_ind

# Load metadata and counts
metadata = pd.read_csv("Diabetes_infection_pheno.csv", index_col=0)
counts = pd.read_csv("Diabetes_infection_count.csv", index_col=0)
print('Counts shape:', counts.shape) # (10346, 397)
print('Metadata shape:', metadata.shape)

# Fix naming mismatch: dots -> dashes
counts.columns = counts.columns.astype(str).str.strip().str.replace('.', '-',
regex=False)
metadata.index = metadata.index.astype(str).str.strip().str.replace('.', '-',
regex=False)

# Verify alignment
print('Samples only in counts:', len(set(counts.columns) - set(metadata.index)))
print('Samples only in metadata:', len(set(metadata.index) - set(counts.columns)))
```

Phase 1 — Sparsity Check & First Filtration (20%)

```
# Sparsity assessment
total_zeros = (counts == 0).sum().sum()
sparsity_pct = (total_zeros / counts.size) * 100
print(f'Sparsity: {sparsity_pct:.2f}%') # 29.74%

# Keep genes expressed in >= 20% of samples
zeros_per_gene = (counts == 0).sum(axis=1)
total_samples = counts.shape[1] # 397
min_samples_required = int(0.2 * total_samples) # 79
max_allowed_zeros = total_samples - min_samples_required # 318

genes_to_keep = zeros_per_gene[zeros_per_gene <= max_allowed_zeros].index
counts_filtered = counts.loc[genes_to_keep]
print(f'Genes before: {counts.shape[0]}') # 10,346
print(f'Genes after: {len(genes_to_keep)}') # 10,119
```

Phase 1 — Sample Grouping & Second Filtration (Mean > 10)

```
# Map days to infection stages
stage_map = {
    'D0': 'Healthy',
    'D1': 'Infection_Early', 'D2': 'Infection_Early',
    'D3': 'Infection_Middle', 'D4': 'Infection_Middle', 'D5': 'Infection_Middle',
    'D7': 'Infection_Late', 'D8': 'Infection_Late',
    'D9': 'Infection_Late', 'D10': 'Infection_Late'
}

metadata['Stage_Group'] = metadata['CL1'].map(stage_map).fillna('Healthy')
print(metadata['Stage_Group'].value_counts())
# Healthy:336 Middle:29 Late:18 Early:14

# Second filtration: mean > 10 in at least one infection stage
early_s = metadata[metadata['Stage_Group']=='Infection_Early'].index
middle_s = metadata[metadata['Stage_Group']=='Infection_Middle'].index
late_s = metadata[metadata['Stage_Group']=='Infection_Late'].index

mean_e = counts_filtered[early_s].mean(axis=1)
mean_m = counts_filtered[middle_s].mean(axis=1)
mean_l = counts_filtered[late_s].mean(axis=1)

keep_mask = (mean_e > 10) | (mean_m > 10) | (mean_l > 10)
counts_final = counts_filtered[keep_mask]
print(f'Final genes: {counts_final.shape[0]}') # 3,262
```

Phase 2 — CPM Normalization, Log2 & PCA

```
# CPM Normalization
counts_cpm = counts_final.div(counts_final.sum(axis=0), axis=1) * 1e6

# Log2 transformation (pseudocount +1 to handle zeros)
counts_log = np.log2(counts_cpm + 1)
print(f'Max: {counts_log.values.max():.2f} Min: {counts_log.values.min():.2f}')

# PCA
from sklearn.decomposition import PCA
pca = PCA(n_components=2)

pca_result = pca.fit_transform(counts_log.T) # samples x genes
print(f'PC1: {pca.explained_variance_ratio_[0]*100:.2f}%') # 25.70%
```

```
print(f'PC2: {pca.explained_variance_ratio_[1]*100:.2f}%') # 13.94%
```

Phase 3 — DEGs: T-test & Biomarker Discovery

```
from scipy import stats

def get_comparison(group_a, group_b):
    idx_a = metadata[metadata['Stage_Group']==group_a].index
    idx_b = metadata[metadata['Stage_Group']==group_b].index
    _, p_vals = stats.ttest_ind(counts_log[idx_a], counts_log[idx_b], axis=1)
    log2fc = counts_log[idx_a].mean(axis=1) - counts_log[idx_b].mean(axis=1)
    return pd.DataFrame({'Gene':counts_log.index, 'p_value':p_vals, 'Log2FC':log2fc})

h_vs_e = get_comparison('Infection_Early', 'Healthy')
h_vs_m = get_comparison('Infection_Middle', 'Healthy')
h_vs_l = get_comparison('Infection_Late', 'Healthy')

# Filter: p < 0.05 and |Log2FC| > 0.58 (~1.5x fold change)
def get_sig(df): return set(df[(df['p_value']<0.05)&(df['Log2FC'].abs())>0.58]['Gene'])
sig_early = get_sig(h_vs_e) # 82 genes
sig_middle = get_sig(h_vs_m) # 142 genes
sig_late = get_sig(h_vs_l) # 22 genes

# Top 20 early biomarkers
top20 = h_vs_e[h_vs_e['Gene'].isin(sig_early)].sort_values('p_value').head(20)
print(top20[['Gene', 'p_value', 'Log2FC']])
```

Phase 4 — Heatmap Visualization

```
import seaborn as sns
import matplotlib.pyplot as plt

# Prepare heatmap data (Top 20 early genes)
heatmap_data = counts_log.loc[list(sig_early)]

# Color-code samples by infection stage
stage_colors = {
    'Healthy': 'green',
    'Infection_Early': 'orange',
    'Infection_Middle': 'red',
    'Infection_Late': 'blue'
}
```

```
sample_colors = metadata['Stage_Group'].map(stage_colors)

# Clustered heatmap
g = sns.clustermap(heatmap_data, cmap='RdYlBu_r', z_score=0,
col_colors=sample_colors,
col_cluster=True, row_cluster=True,
xticklabels=False, yticklabels=True,
figsize=(12, 10))
g.fig.suptitle('Top 20 Early Detection Biomarkers Heatmap', fontsize=16)
plt.show()
```

Esraa Ramadan Mohamed Ahmed

aass20201011@gmail.com | Master's Student in Molecular Medicine / Biomedical Sciences

Objective

Motivated master's student in Molecular Medicine with hands-on experience in RNA-seq bioinformatics, multi-omics data analysis, and computational biology. Seeking to contribute to cutting-edge genomic research with a strong foundation in both wet-lab biology and data science.

Education

Master's Student — Molecular Medicine / Biomedical Sciences

Currently enrolled | Focus: Computational Biology, Genomics, Epigenetics

Research Experience

Research Project — RNA-seq Bioinformatics Pipeline

- Developed a complete RNA-seq bioinformatics pipeline from raw counts to biomarker discovery.
- Analyzed 10,346 genes across 397 samples from a real-world Diabetes & Infection dataset.
- Performed quality control, CPM normalization, Log2 transformation, and PCA.
- Identified 82 Early, 142 Middle, and 22 Late differentially expressed genes (DEGs) using T-test.
- Discovered a robust 20-gene biomarker signature with Steady and Switching marker classes.
- Validated results via hierarchical clustering heatmaps and Venn diagram cross-validation.

Technical Skills

Programming	Python, R
Bioinformatics Tools	DESeq2, MOFA2, T-test pipelines, PCA, Heatmap, Venn Diagram
Data Analysis	RNA-seq, Multi-omics, Gene Expression Analysis, Biomarker Discovery
Visualization	ggplot2, pheatmap, matplotlib, seaborn
Research Areas	Molecular Medicine, Epigenetics, Spatial Transcriptomics, Cancer Genomics

Project Repository

GitHub: <https://github.com/aass20201011-lab/RNA-seq-Project> — RNA-seq Bioinformatics Pipeline for Early Infection Detection

Application Email — Ready to Send

To: admissions@zewailcity.edu.eg / [Research Lab Contact]

Subject: MSc Application — RNA-seq Bioinformatics Research Background | Esraa Ramadan

From: aass20201011@gmail.com

Dear Admissions Committee / [Professor's Name],

My name is Esraa Ramadan Mohamed Ahmed, and I am writing to express my strong interest in joining the MSc program in Biomedical Sciences at Zewail City of Science and Technology.

I recently completed a bioinformatics research project in which I developed a full RNA-seq analysis pipeline applied to a real-world Diabetes & Infection dataset (10,346 genes, 397 samples). The project involved data preprocessing, CPM normalization, PCA, differential expression analysis (T-test & DESeq2), and the identification of a 20-gene early infection biomarker signature — classifying markers into Steady (always-on) and Switching (molecular timer) categories.

This work deepened my interest in computational genomics and confirmed my goal of pursuing graduate-level research in this field. I believe Zewail City's MSc program — particularly the Computational Biology concentration — is the ideal environment for me to grow as a researcher.

I have attached my CV and included a link to my project repository below.

GitHub: <https://github.com/aass20201011-lab/RNA-seq-Project>

I would be grateful for the opportunity to discuss my background further.

Sincerely,
Esraa Ramadan Mohamed Ahmed
aass20201011@gmail.com